

Q1. (b) 2

Q2. $2x^2 - 7x + 5 = 0$

$$2x^2 - 2x - 5x + 5 = 0$$

$$2x(x - 1) - 5(x - 1) = 0$$

$$(2x - 5)(x - 1) = 0$$

$$x = 5/2, x = 1$$

Q3. $a = 3, d = 7 - 3 = 4, n = 20$

$$S_n = n/2 [2a + (n - 1)d]$$

$$S_{20} = 20/2 [2 \times 3 + 19 \times 4]$$

$$= 10 [6 + 72]$$

$$= 10 \times 78 = 780$$

Q4. $\sin A = 3/5$

$$\cos A = 1 - \sin A = 1 - 3/5 = 2/5$$

$$\tan A = \sin A / \cos A = (3/5) / (2/5) = 3/2$$

$$\sec A = 1 / \cos A = 5/2$$

$$\begin{aligned} \text{Q5. } 1/R &= 1/2 + 1/3 + 1/6 \\ &= (3 + 2 + 1)/6 = 6/6 = 1 \end{aligned}$$

$$R = 1$$

$$I = 6/1 = 6$$

Q6. Don't know



It is a redox reaction.

Q8. The left ventricle pumps oxygenated blood to all parts of the body, so it has to pump with more force. The right ventricle pumps blood only to the lungs. So the left wall is thicker.

Q9. (i) Total outcomes = 36

Sum 7 : (1,6) (2,5) (3,4) (4,3) (5,2)

$$P(\text{sum } 7) = 5/36$$

(ii) Total balls = 5 + 3 = 8

$$P(\text{blue}) = 3/8$$

Q10. $u = 20 \text{ cm}$, $f = -15 \text{ cm}$

$$1/v + 1/u = 1/f$$

$$1/v = 1/f - 1/u = -1/15 - 1/20$$

$$= (-4 - 3)/60 = -7/60$$

$$v = -60/7 = -8.57 \text{ cm}$$

Image is 8.57 cm in front of the mirror.